

Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

IMCQ 5

St. George's University

1. A 23-year-old man is brought to the emergency department after being involved in a motor vehicle collision 20 minutes ago. Neurologic examination and imaging studies are consistent with a right hemisection of the spinal cord at the T9. Which of the following deficits is most likely present in this patient?

- A. Paralysis of right upper limb
- B. Paralysis of left upper limb
- C. Loss of pain sensation in right upper limb
- D. Loss of vibration sensation in left upper limb
- E. Loss of pain sensation in right lower limb
- F. Loss of vibration in left lower limb
- G. Loss of thermoception in left lower limb
- H. Loss of touch sensation in left lower limb

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- E. Loss of pain sensation in right lower limb
- F. Loss of vibration in left lower limb
-  G. Loss of thermoception in left lower limb
- H. Loss of touch sensation in left lower limb

2. A 50-year-old man comes to the physician because of a 2-day history of numbness on the right side of his face. Neurologic examination shows loss of pain sensation on his right face. Which of the following tracts is most likely affected in this patient?

- A. Left anterolateral system
- B. Right anterolateral system
- C. Left spinothalamic
- D. Right spinothalamic
- E. Left spinal trigeminal
- F. Right spinal trigeminal

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3. A 65-year-old man comes to the physician because of a 2-week history of numbness. He has type 2 diabetes and is on daily oral hypoglycemics. His vital signs are within normal limits. Neurologic examination is consistent with a defect in the dorsal column-medial lemniscus system. Which of the following somatosensory modalities are most likely affected in this patient?

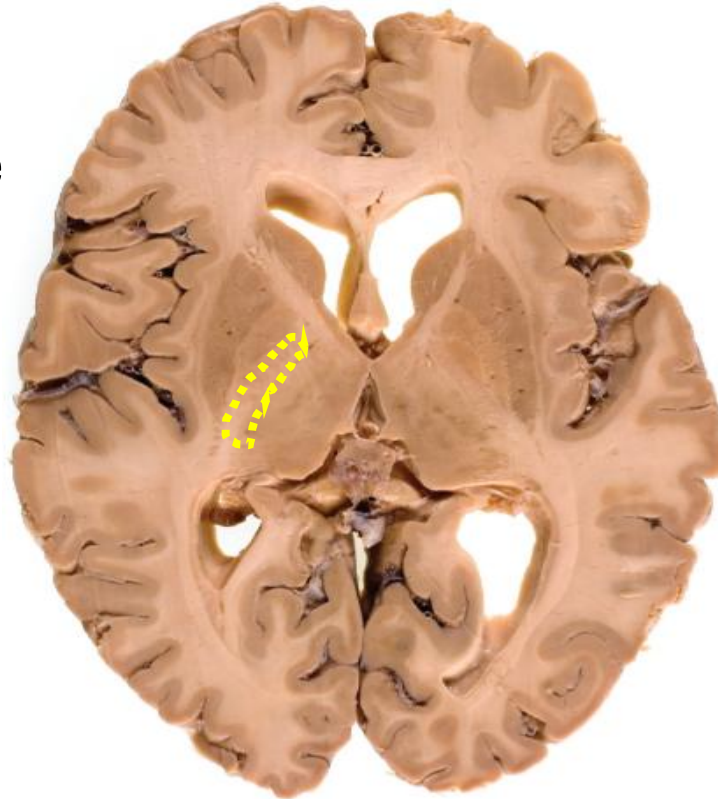
- A. Pain and proprioception
- B. Discriminative touch and pain
- C. Pain and temperature
- D. Temperature and discriminative touch
- E. Vibration and proprioception

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4. A 76-year-old woman comes to the physician because of a 1-day history of unilateral weakness. She has a history of hypertension and is non-adherent with her medications. Her blood pressure is 160/100 mm Hg; other vital signs are within normal limits. Neurologic examination shows several abnormalities. Imaging shows a lesion in the area marked on the photograph of a brain section below. Which of the following additional neurologic findings is most likely in this patient?

- A. Extensor plantar response
- B. Facial paralysis
- C. Fibrillations
- D. Hyporeflexia
- E. Dysphagia



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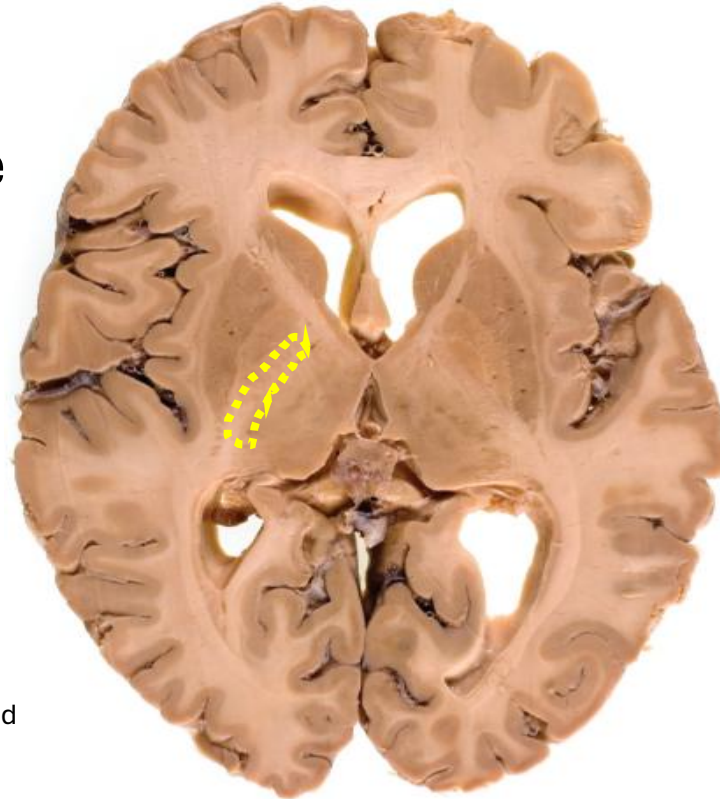
A. Extensor plantar response

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C. Fibrillations

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SOM.MK.I.BPM2.1.NB.1.NSCI.0149 Describe the origin, route, and termination of the lateral corticospinal pathway

SOM.MK.IV.BPM2.4.NB.2.NSCI.0141 Explain the normal and abnormal plantar reflex

SOM.MK.II.BPM2.4.NB.2.NSCI.0152 Contrast common manifestations of lower versus upper motor neuron lesions

5. A 43-year-old man is brought to the emergency department after being involved in a motor vehicle collision 30 minutes ago. Neurologic examination and imaging studies are consistent with a left hemisection of the spinal cord at the C5 vertebral level. Which of the following limbs is/are most likely paralyzed in this patient?

- A. Both upper limbs
- B. Both lower limbs
- C. Right upper limb
- D. Left lower limb
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6. A 24-year-old man comes to the physician because of a 1-week history of numbness and tingling in his right arm. Vital signs are within normal limits. Neurologic examination shows weakness on extension of his right elbow. There is also a diminished triceps reflex on the right. Which of the following spinal levels is most likely affected in this patient?

- A. C5, C6
- B. C7, C8
- C. L2, L3
- D. L5, S1
- E. T8, T9

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7. A 22-year-old woman comes to the physician because of a 3-month history of bilateral tingling of her hands, arms, neck, and upper trunk. Two years ago, she sustained a traumatic cervical injury. Neurologic examination shows a bilateral loss of nociception and thermoception for dermatomes C4 and C5. Proprioception, vibration sense, touch, and motor function are spared. Which of the following structures is most likely damaged?

- A. Bilateral thalamus
- B. Bilateral hypothalamus
- C. Central spinal cord (C5 – C6)
- D. Central spinal cord (C2 – C7)
- E. Left spinal cord (C5 – C6)
- F. Right spinal cord (C2 – T2)
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8. A 46-year-old man comes to the emergency department because of 3-hour history of weakness affecting his right leg. Neurologic examination shows exaggerated stretch reflexes of his right leg. The application of a noxious stimulus to his right foot provokes ipsilateral plantar extension. Which of the following neuronal fiber types most likely contributes to the afferent limb of the plantar reflex?

- A. Ia
- B. Ib
- C. II
- D. III
- E. IV

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 D. III

E. IV

9. A 46-year-old woman comes to the physician because of a 12-hour history of redness and pain in her right eye. Examination shows redness of the right conjunctiva consistent with conjunctivitis. Vision and eye movements are normal. Which of the areas marked in the images below best describes the entry point of the cranial nerve carrying the pain sensations from the affected structure into the brainstem?

A. A

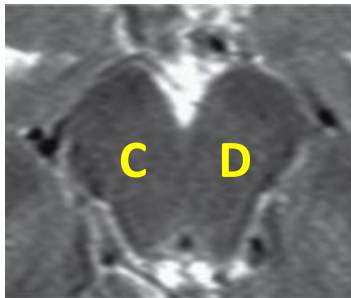
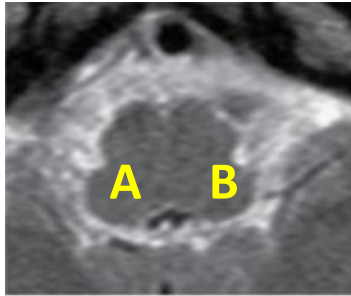
B. B

C. C

D. D

E. E

F. F



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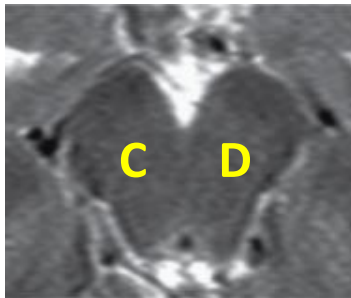
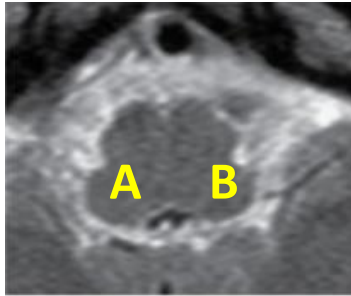
B. B

C. C

D. D

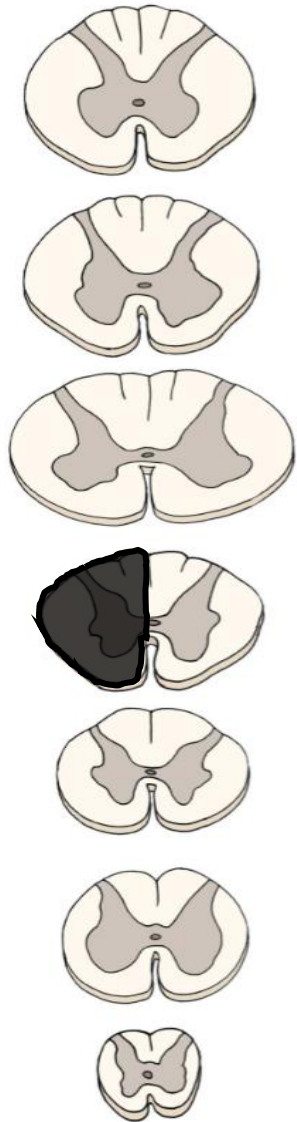
😊 E. E

F. F



SOM.MK.I.BPM2.1.NB.1.NSCI.0121 describe the complete pathways for pain and temperature, including the localization in different sections of the central nervous system
SOM.MK.I.BPM2.4.NB.4.NSCI.0122 use the knowledge of pain and temperature pathways in a clinical scenario to localize a lesion

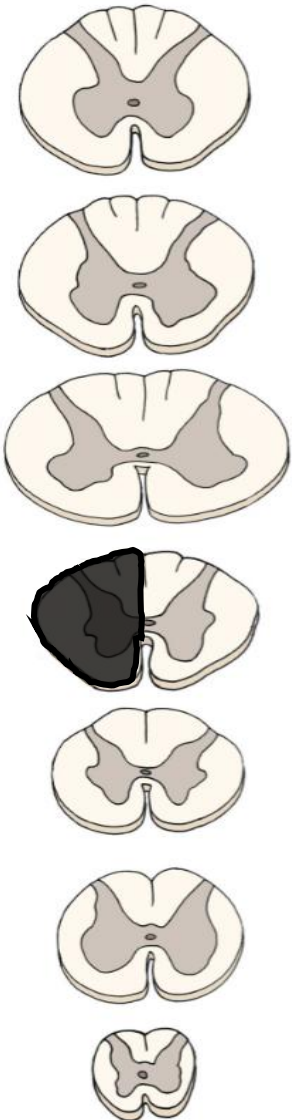
Right	Left
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10. A 24-year-old man is brought to the emergency department by paramedics 30 minutes after he was involved in a motor vehicle collision. Examination and imaging studies are consistent with the lesion shown in the diagram. Which of the following extremities is/are most likely affected by losses of pain sensations in this patient?

- A. Left lower
- B. Left upper and lower
- C. Left upper
- D. Right lower
- E. Right upper and lower
- F. Right upper limb

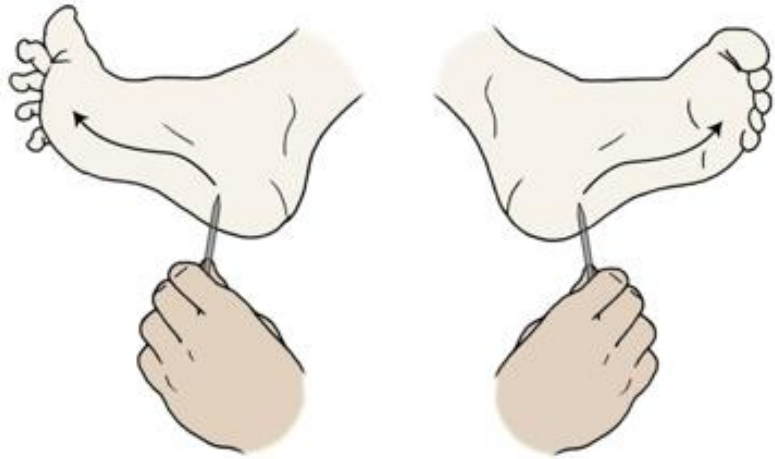
Right	Left
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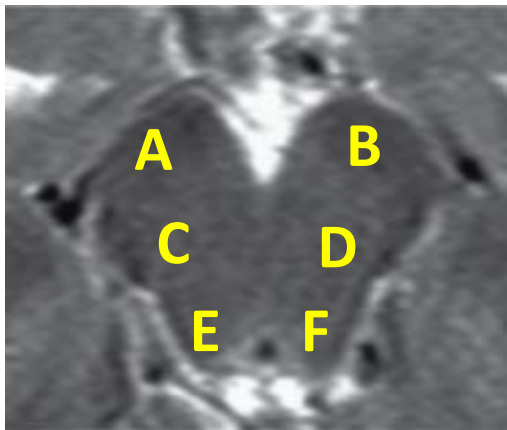
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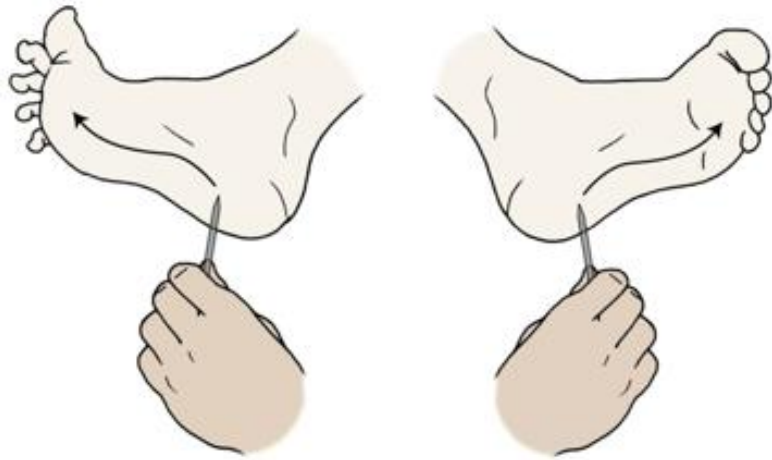
11. A 41-year-old woman is brought to the emergency department because of the sudden onset of severe motor deficits. A complete neurologic examination is performed. The outcome of plantar response testing is shown in the schematic drawing below. Which of the labels on the clinical image of a horizontal CNS section best represents the location of the lesion?



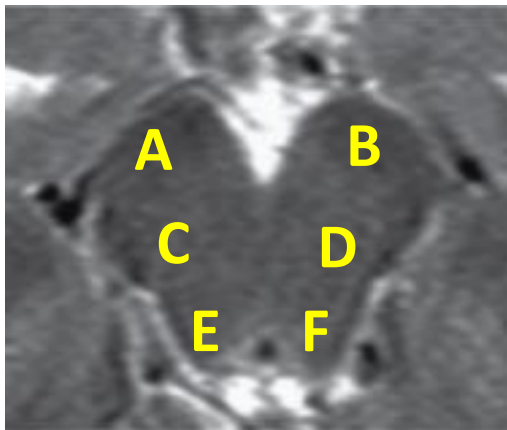
- A. A
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- C. C
- D. D
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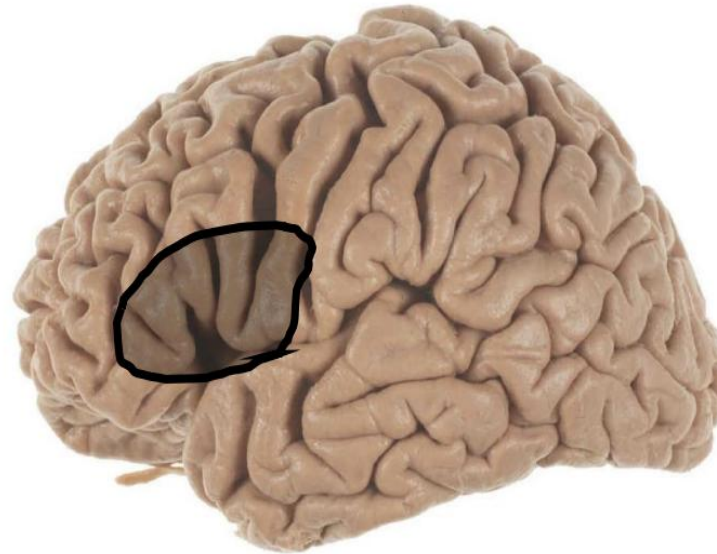
- A. A
- 😊 B. B
- C. C
- D. D
- E. E
- F. F



SOM.MK.IV.BPM2.4.NB.2.NSCI.0141 Explain the normal and abnormal plantar reflex
 SOM.MK.I.BPM2.4.NB.2.NSCI.0153 Correlate upper and lower motor neuronal signs with lesions affecting central and peripheral nervous site

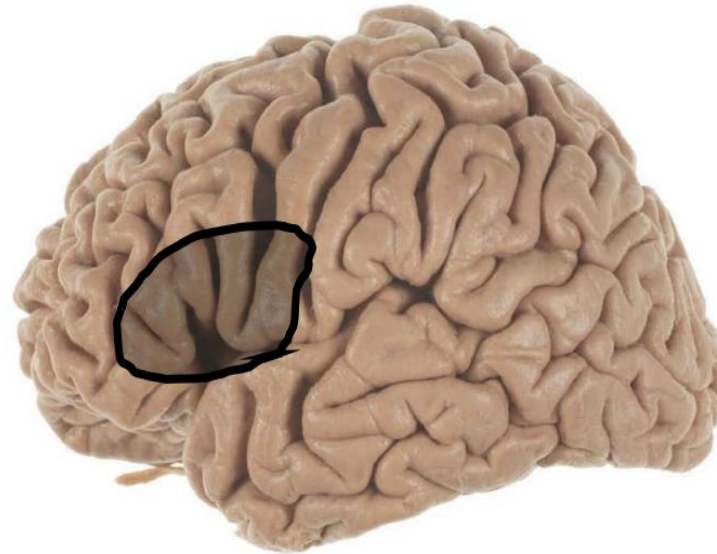
12. A 29-year-old man comes to the physician because of a 1-day history of weakness of his right upper extremity and a distorted face. The right corner of his mouth is drooping, and the right nasolabial fold is flat. He has difficulty speaking. Imaging shows a lesion in the area marked on the picture of a cerebral hemisphere below. Which of the following additional findings is most likely expected in this patient?

- A. Diplopia
- B. Inability to wrinkle forehead on the right
- C. Inability to wrinkle forehead on the left
- D. Impaired language comprehension
- E. Deviation of the tongue to the right on protrusion
- F. Deviation of the tongue to the left on protrusion



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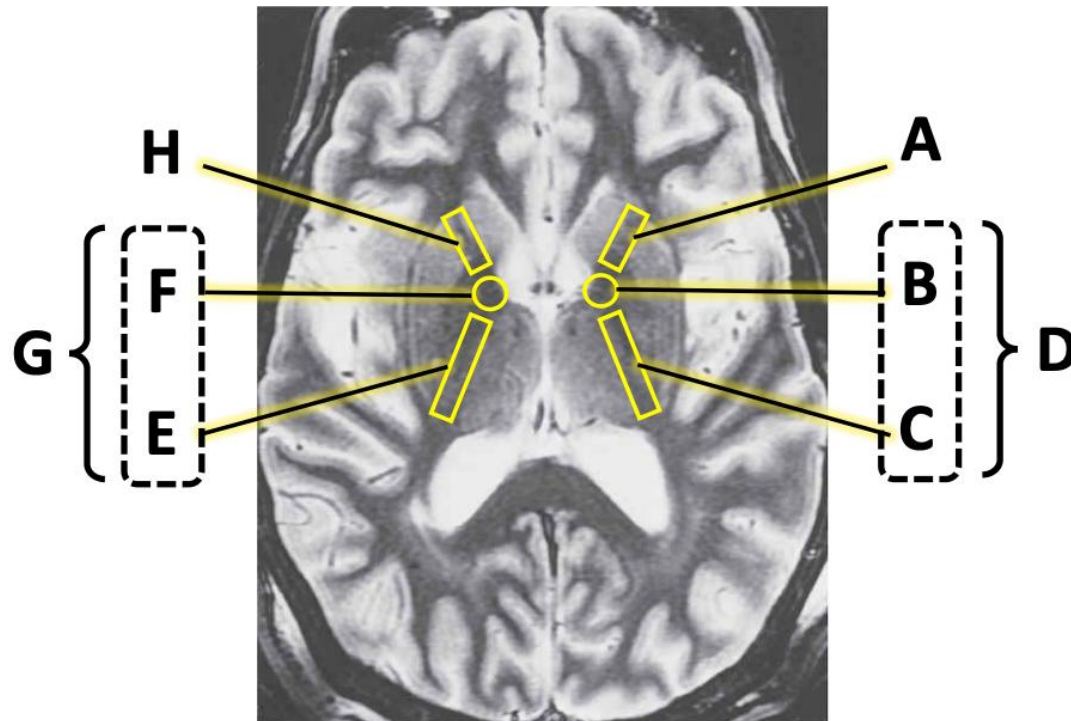
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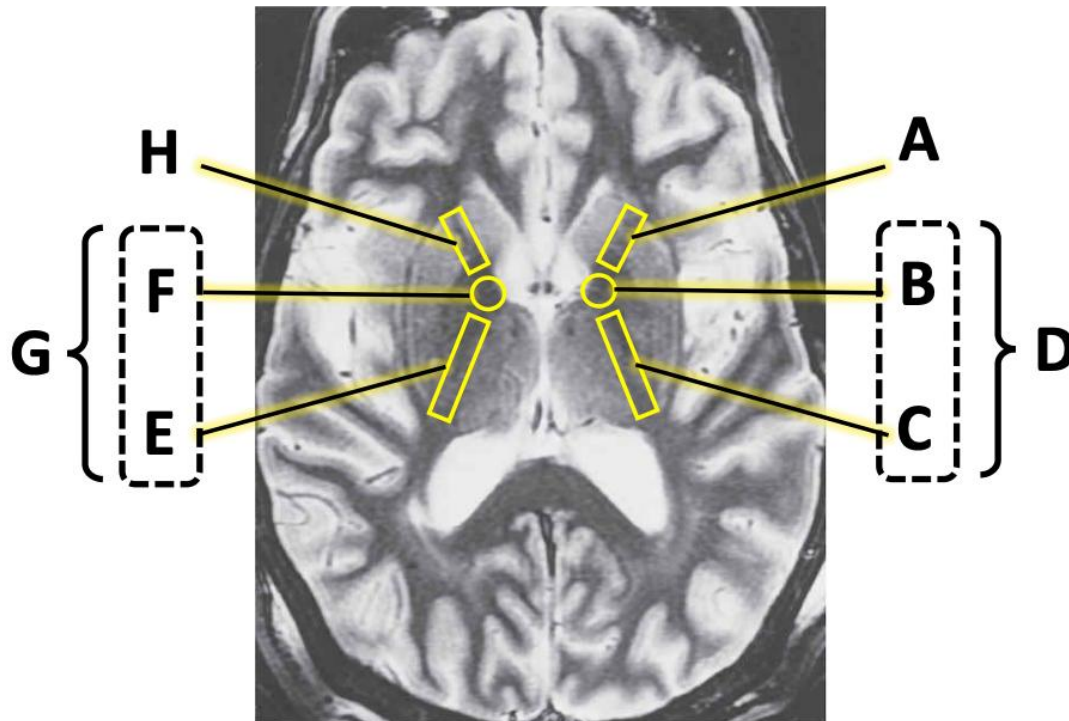
13. A 41-year-old woman is brought to the emergency department because of a sudden onset of severe motor deficits. Neurologic examination shows spastic paralysis and hyperreflexia of the right upper and lower extremities. Wrinkling the forehead and smiling are normal bilaterally. A Babinski sign is present on the right. Touch, vibration, proprioception, and pain and temperature sensations are absent on the right body and face. In which of the areas labeled on the MRI of a normal brain is the lesion most likely located?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H



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- A. A
 B. B
 😊 C. C
 D. D
 E. E
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 G. G
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SOM.MK.I.BPM2.4.NB.2.NSCI.0153 Correlate upper and lower motor neuronal signs with lesions affecting central and peripheral nervous site

14. A 45-year-old woman comes to the physician because of a 3-day history of difficulty walking. Physical examination findings and laboratory studies are consistent with Guillain-Barré syndrome. Which of the following structures is most likely affected in this patient?

- A. End plate
- B. Nerve ending
- C. Schwann cells
- D. Soma
- E. Muscle fiber

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The End of BPM2 NB IMCQ 5

If you have any questions, please send an email to:

Neurosciences ASK ASK-NSCI@sgu.edu